

#	Asset	Classification	Threat	Risk	Risk Level	Recommended Control
1	Student Information System (SIS)	Software	Unauthorized access / account compromise	Exposure or modification of student records	High	Multi-Factor Authentication (MFA)
2	University Database Server	Hardware	Ransomware or unauthorized access	Loss or theft of critical university data	High	Regular encrypted backups
3	Student & Staff Records	Data	Data breach / insider threat	Exposure of sensitive personal information	High	Encryption and access control
4	University Email System	Service	Phishing and credential theft	Compromised accounts and further attacks	High	MFA + email security filtering
5	Administrator Accounts	Identity	Password theft / privilege abuse	Attacker gains administrative control	High	Privileged Access Management (PAM)
6	Campus Wi-Fi Network	Service	Unauthorized network access	Attackers access internal university resources	Medium	WPA3 and network segmentation
7	Faculty Laptops	Hardware	Malware / device theft	Loss of data and unauthorized access	Medium	Endpoint protection and full-disk encryption
8	Learning Management System (LMS)	Software	Web application attack	Disruption of online learning and data exposure	Medium	Web Application Firewall (WAF)

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9	University Website	Software	DDoS / website compromise	Website becomes unavailable or is defaced	Medium	DDoS protection and monitoring
10	Backup Storage	Data	Ransomware / deletion	Inability to recover important university data	High	Offline/immutable backups
11	Student ID Cards	Identity	Identity theft / card misuse	Unauthorized access to campus facilities	Low	ID verification and access logging
12	University Computer Lab PCs	Hardware	Malware / unauthorized software	Data loss and disruption of lab activities	Low	Antivirus and application restrictions

Answers to Student Questions

6. Which asset is most critical?

The **University Database Server** is the most critical asset because it stores and provides access to large amounts of essential university information, including student and staff records.

7. Which risk has the greatest impact?

The greatest-impact risk is **unauthorized access or ransomware affecting the university database**, because it could cause data loss, privacy violations, financial damage, and major disruption to university operations.

8. Which control is preventive and which is detective?

- **Preventive control: Multi-Factor Authentication (MFA)** — prevents unauthorized users from accessing accounts even if a password is stolen.
- **Detective control: Security monitoring and logging** — detects suspicious activities and unauthorized access after or while they occur.
- **Backup control:** Encrypted/offline backups are primarily **recovery/corrective controls**, because they help restore data after an incident.

Five Highest Risks

The five **High-risk** assets that should receive priority are:

1. **Student Information System** → MFA
2. **University Database Server** → Encrypted regular backups
3. **Student & Staff Records** → Encryption + access control
4. **University Email System** → MFA + email filtering
5. **Administrator Accounts** → Privileged Access Management (PAM)